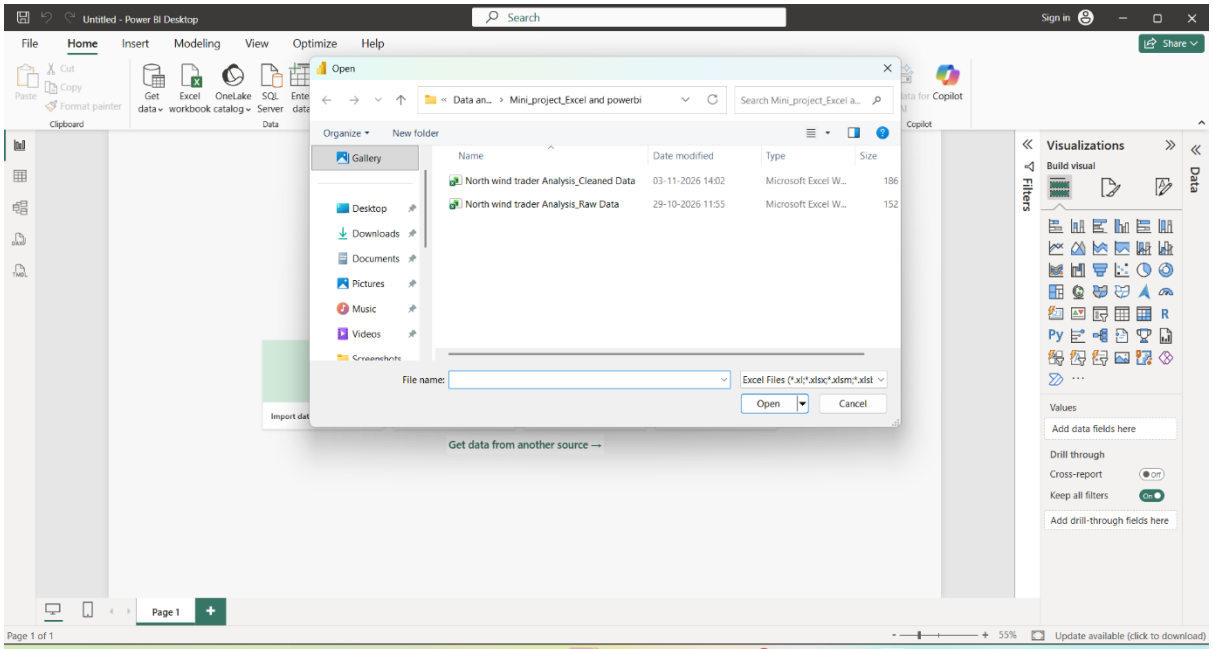
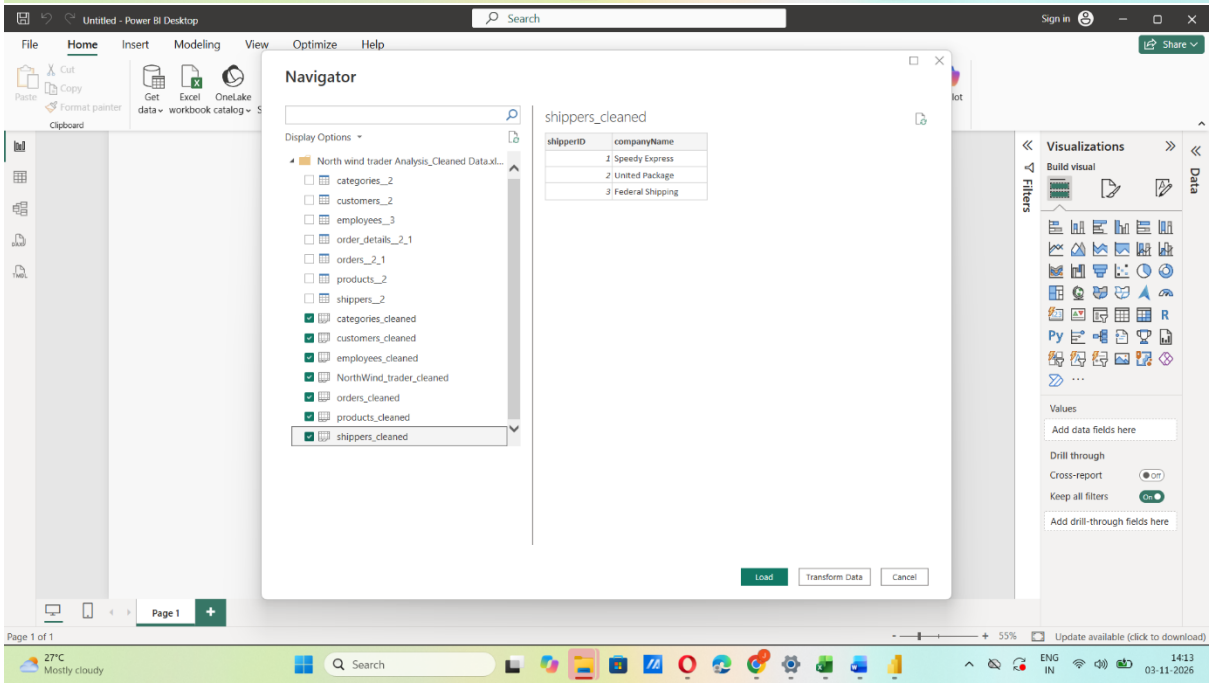
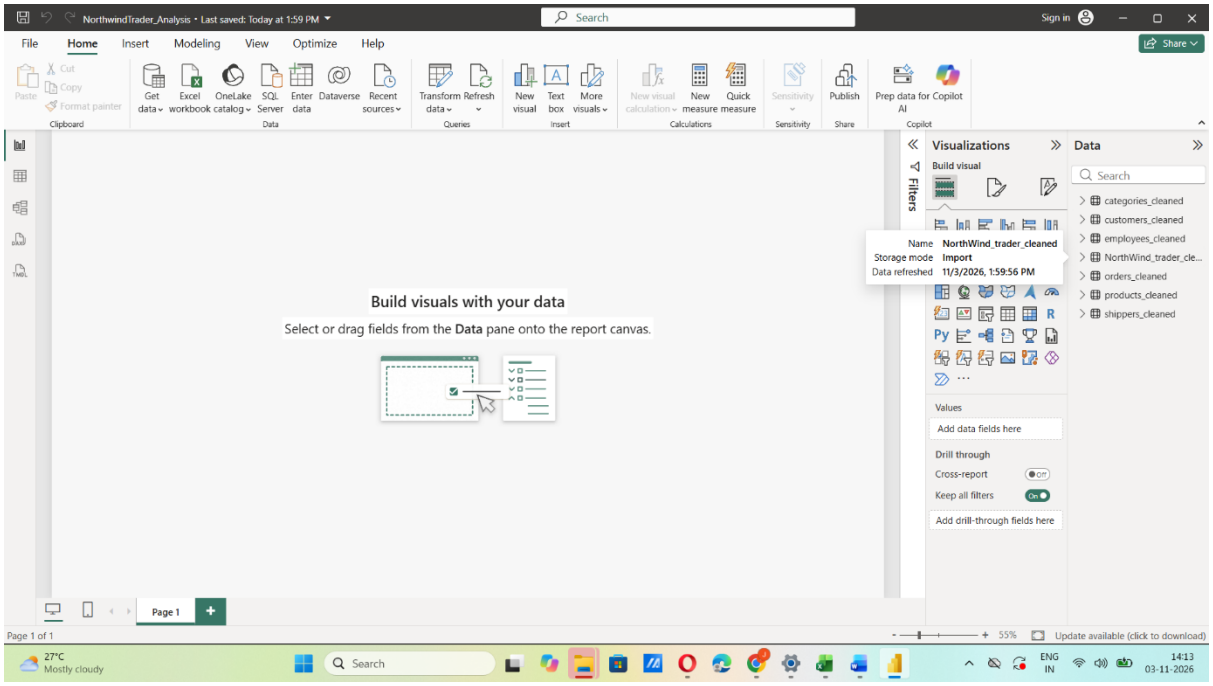
NorthWind Traders Analysis PowerBI Observation

Step 1: imported my Cleaned data into PowerBI workspace.



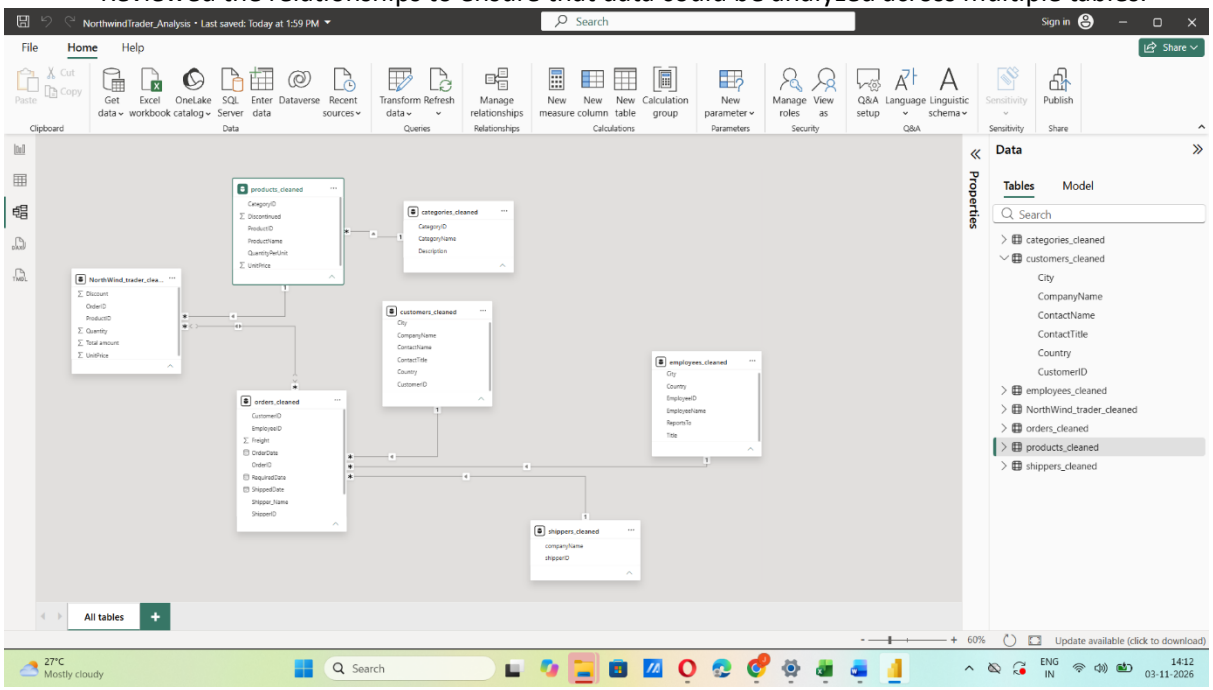


Data is loaded and visible in right side of the Application.



Step 2: Data Model View

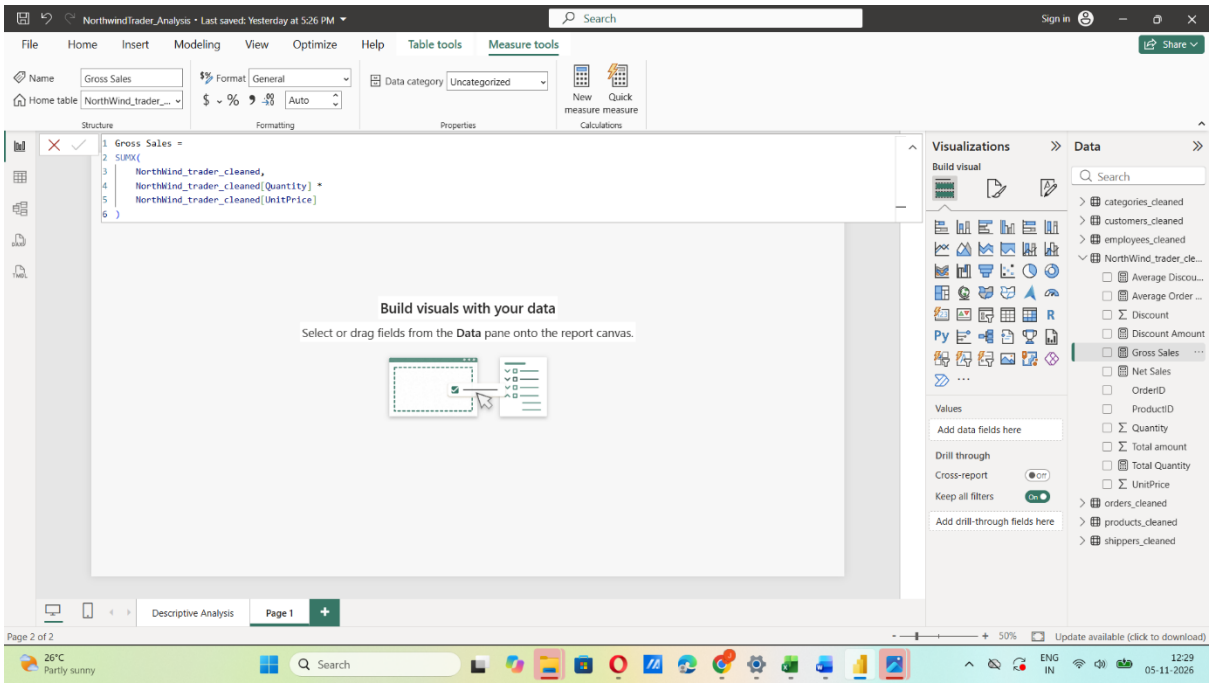
- Imported the cleaned tables into Power BI.
- Opened Model View to examine the relationships between the tables.
- Identified common columns such as Customer ID, Order ID, Product ID, Employee ID and Category ID.
- Used One-to-Many (1:*) relationships wherever applicable.
- Reviewed the relationships to ensure that data could be analyzed across multiple tables.



3.Dax calculation.

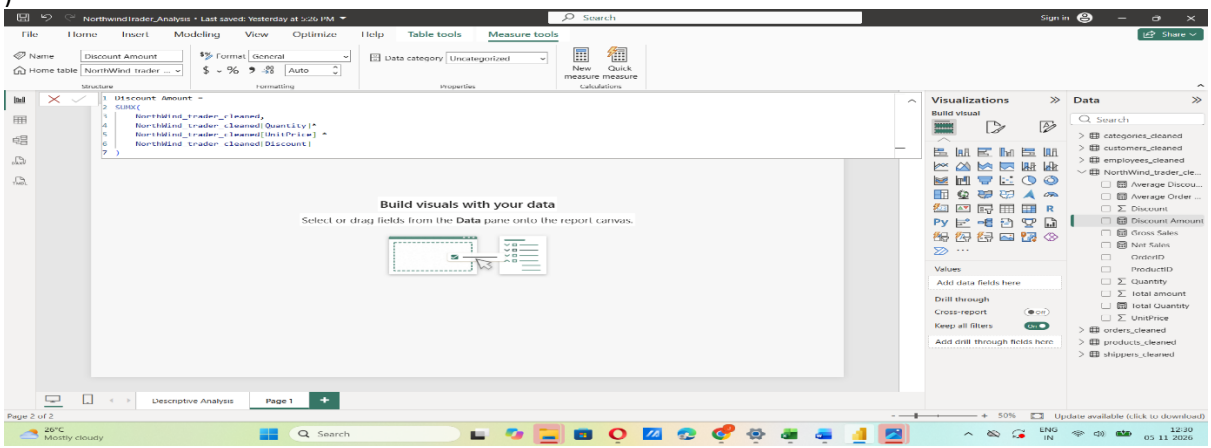
1. Gross sale calculation

```
Gross Sales =
SUMX(
   NorthWind_trader_cleaned,
   NorthWind_trader_cleaned[Quantity] *
   NorthWind_trader_cleaned[UnitPrice]
)
```



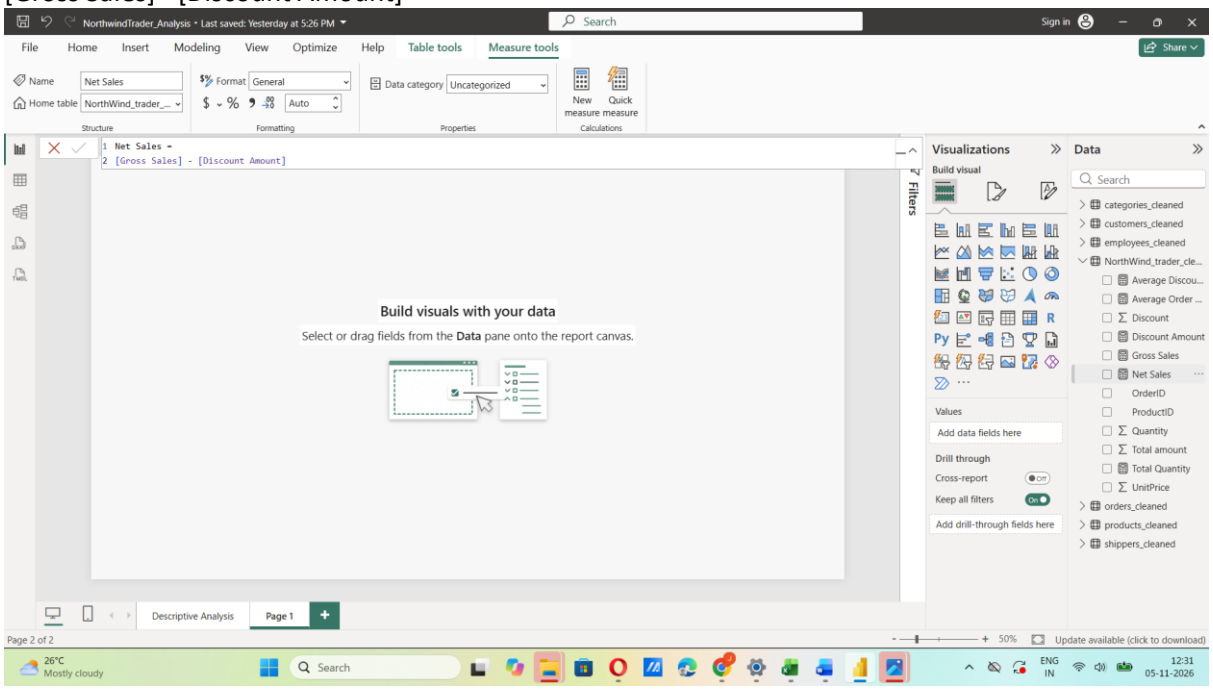
2.Discount Amount

```
Discount Amount =
SUMX(
   NorthWind_trader_cleaned,
   NorthWind_trader_cleaned[Quantity]*
   NorthWind_trader_cleaned[UnitPrice] *
   NorthWind_trader_cleaned[Discount]
)
```



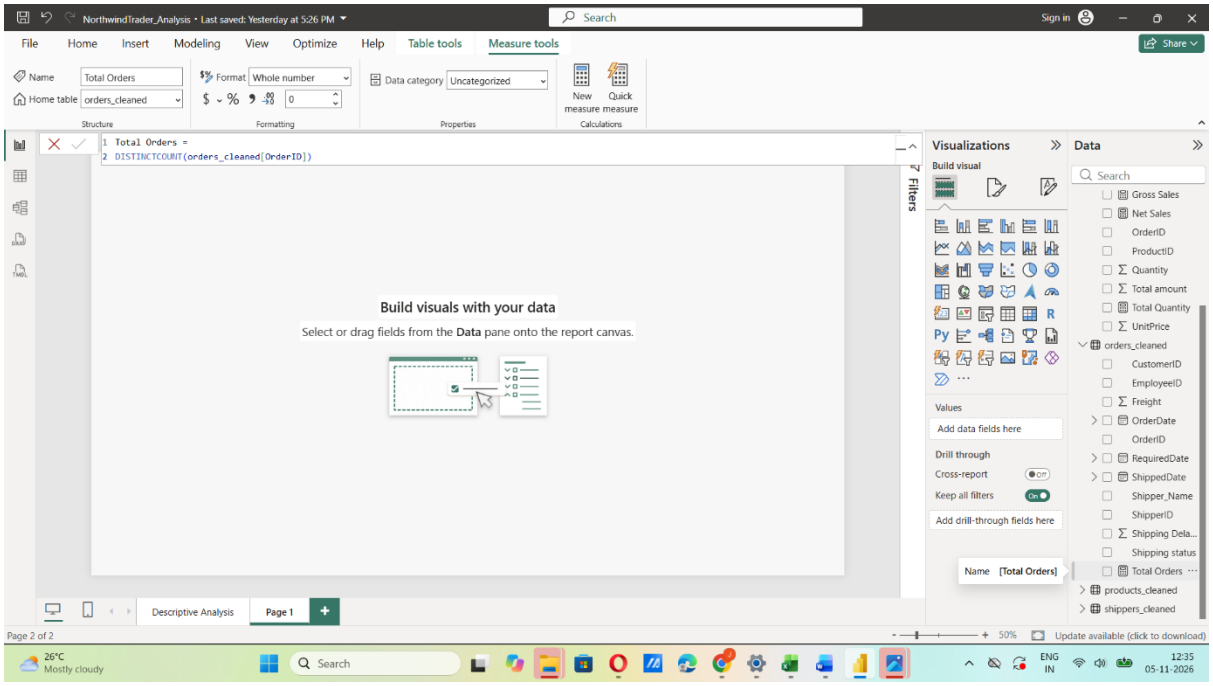
3. **Net sales calculation**

Net Sales =
[Gross Sales] - [Discount Amount]



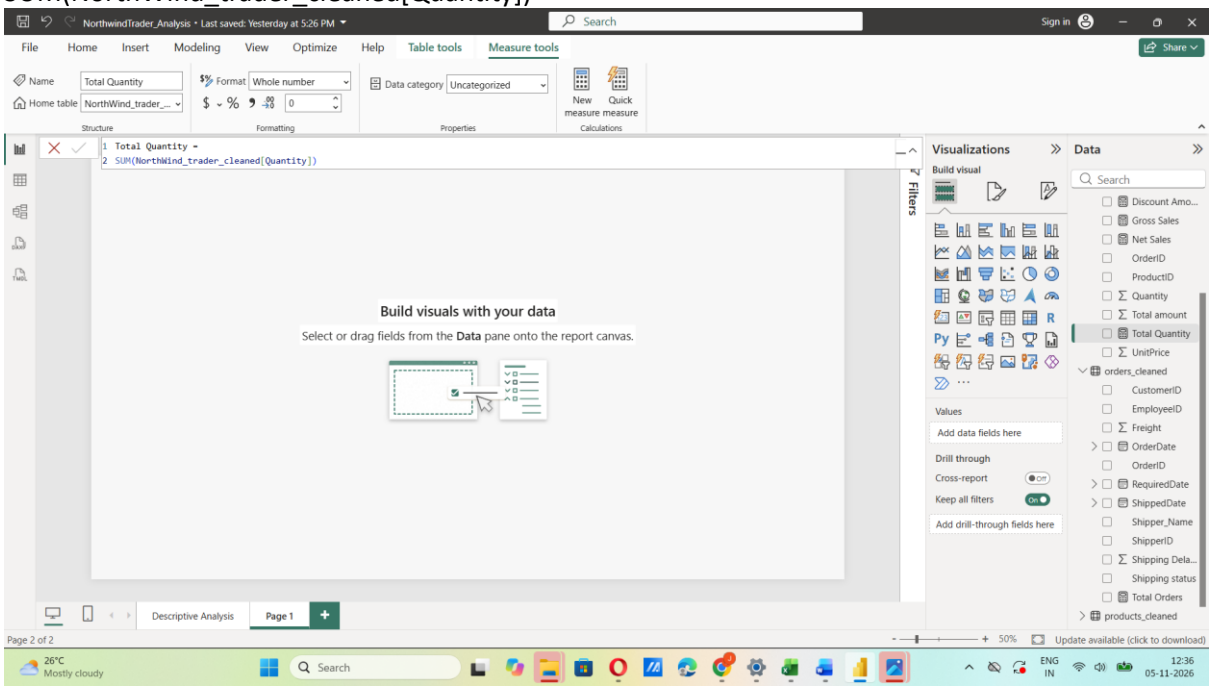
4.Total order

Total Orders =
DISTINCTCOUNT(orders_cleaned[OrderID])



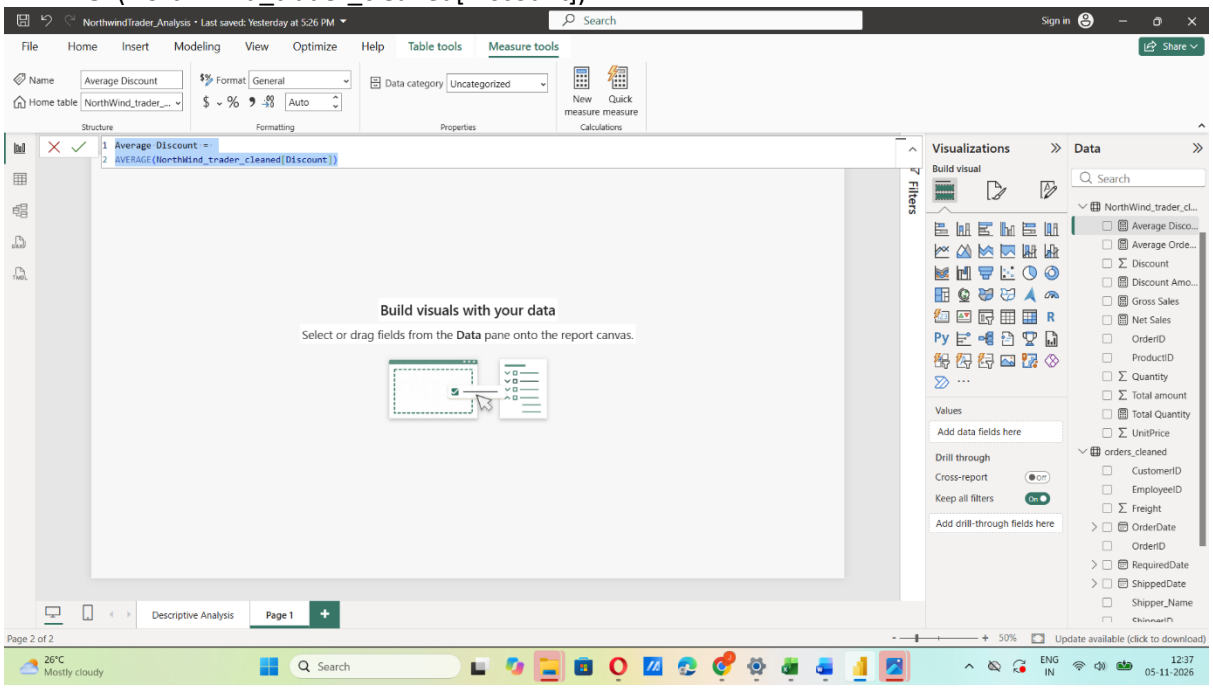
5. Total quantity

Total Quantity =
SUM(NorthWind_trader_cleaned[Quantity])

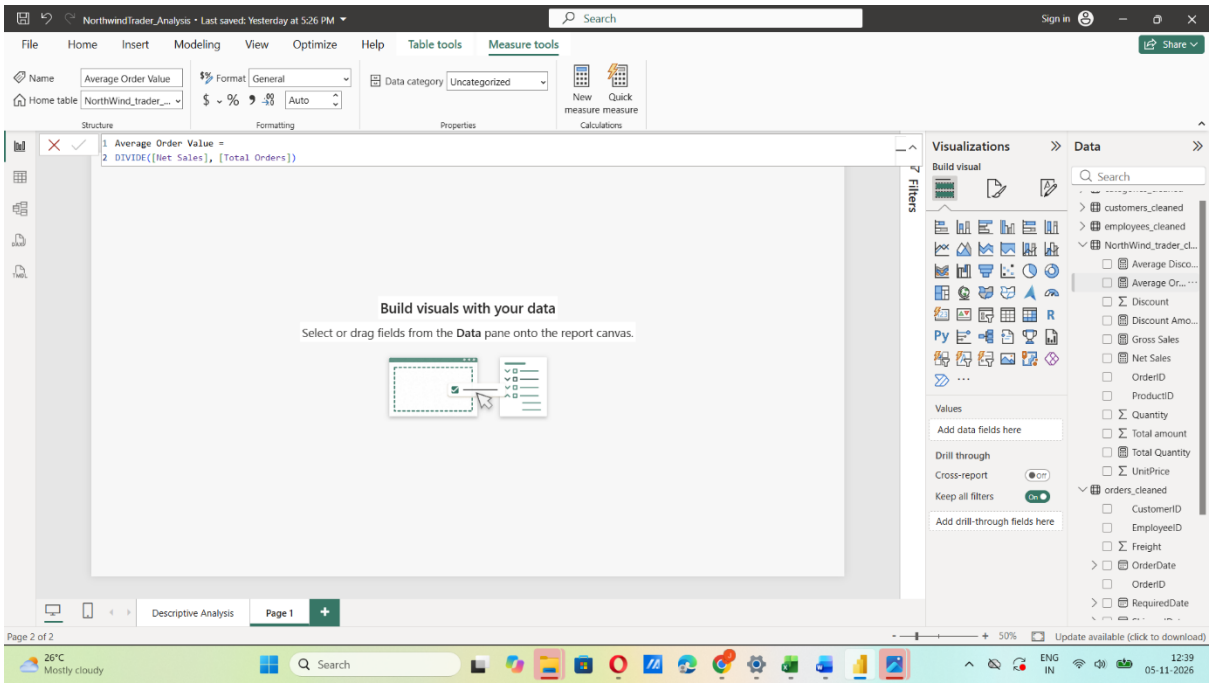


6. **Average Discount value**

Average Discount =
AVERAGE(NorthWind_trader_cleaned[Discount])

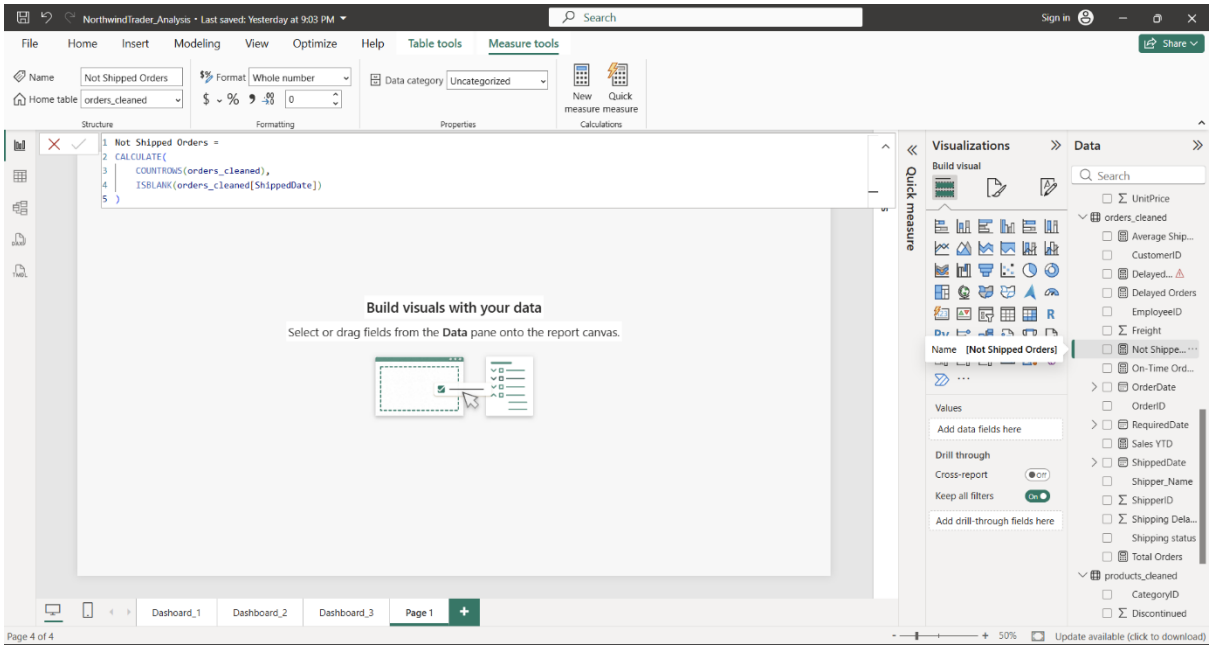


7.**Average order value calculation**
Average Order Value =
DIVIDE([Net Sales], [Total Orders])



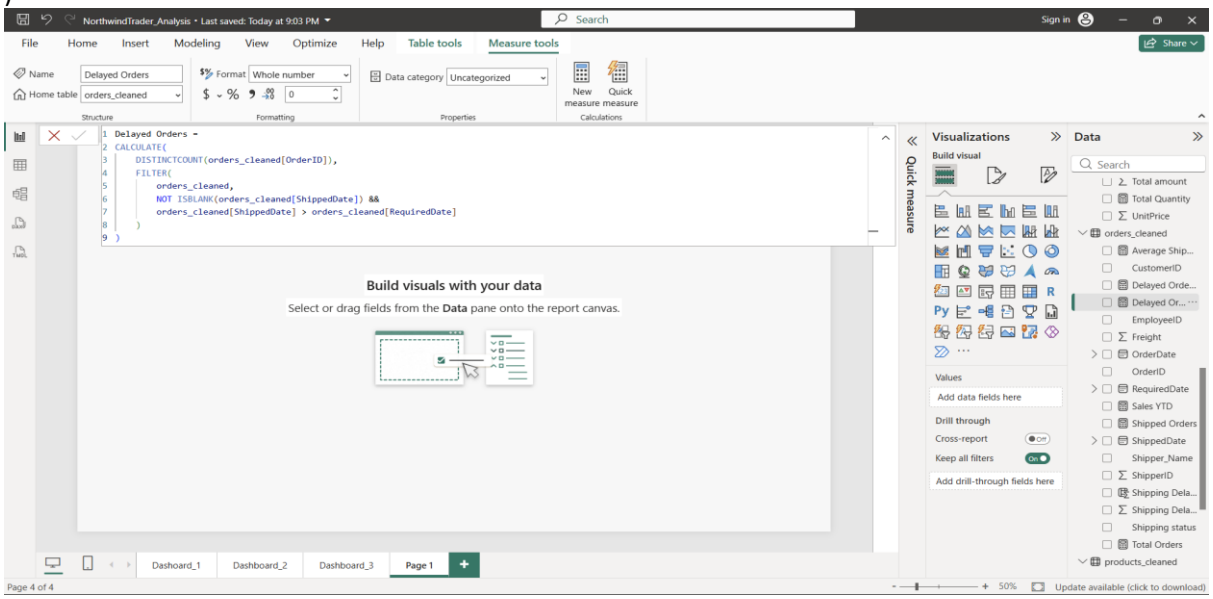
8. created a measure to calculate not shipped orders.

Not Shipped Orders =
CALCULATE(
  COUNTROWS(orders_cleaned),
  ISBLANK(orders_cleaned[ShippedDate])
)



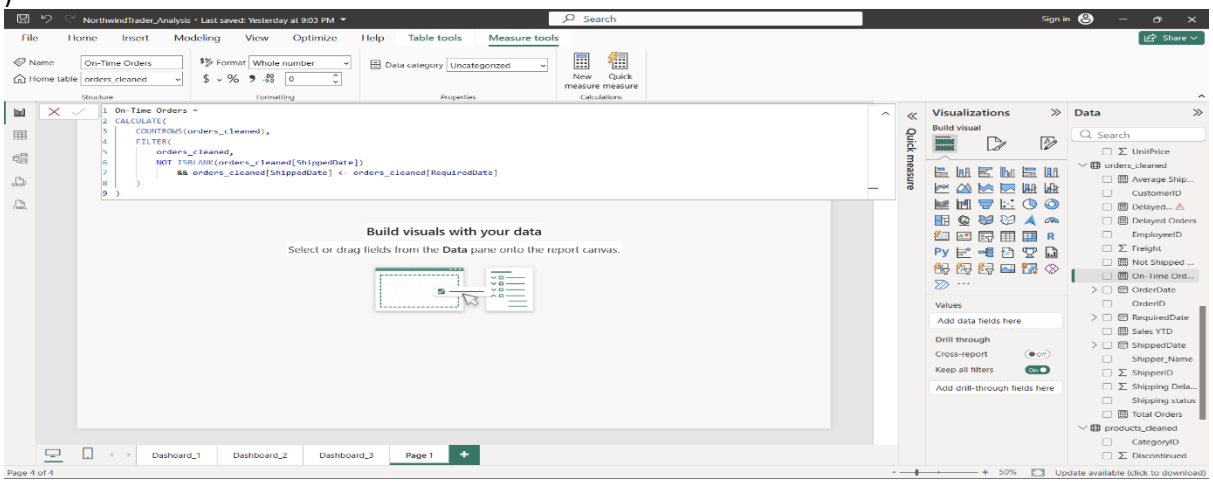
9. Created measure for delayed order

```
Delayed Orders =
CALCULATE(
   DISTINCTCOUNT(orders_cleaned[OrderID]),
   FILTER(
      orders_cleaned,
      NOT ISBLANK(orders_cleaned[ShippedDate]) &&
      orders_cleaned[ShippedDate] > orders_cleaned[RequiredDate]
   )
)
```



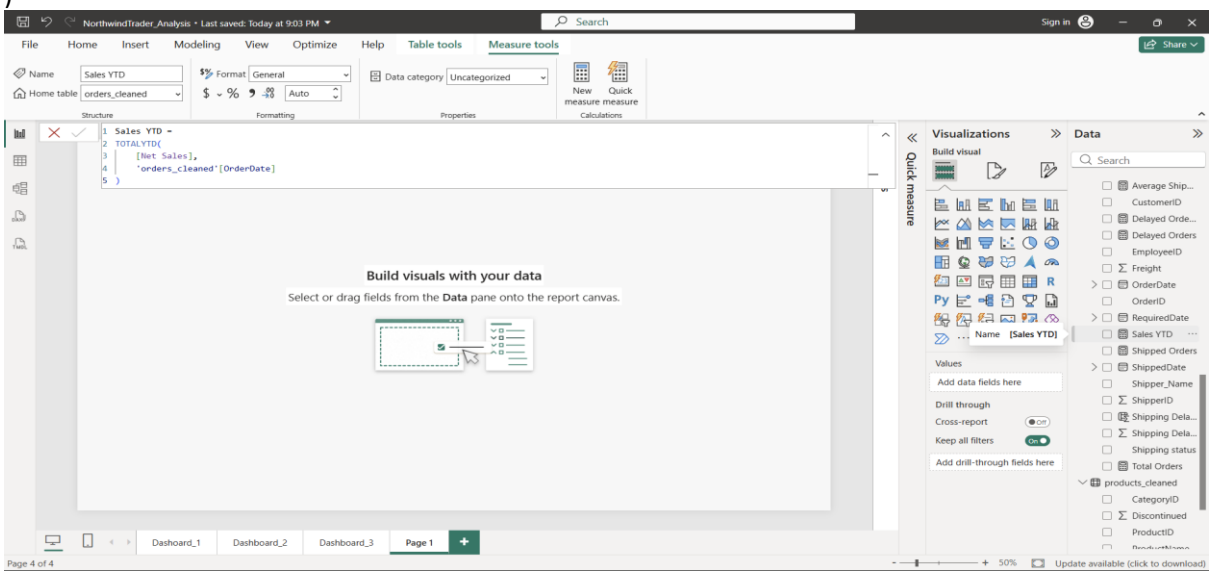
10.1created measure for on time shipped orders.

```
On-Time Orders =
CALCULATE(
   COUNTROWS(orders_cleaned),
   FILTER(
      orders_cleaned,
      NOT ISBLANK(orders_cleaned[ShippedDate])
         && orders_cleaned[ShippedDate] <= orders_cleaned[RequiredDate]
   )
)
```



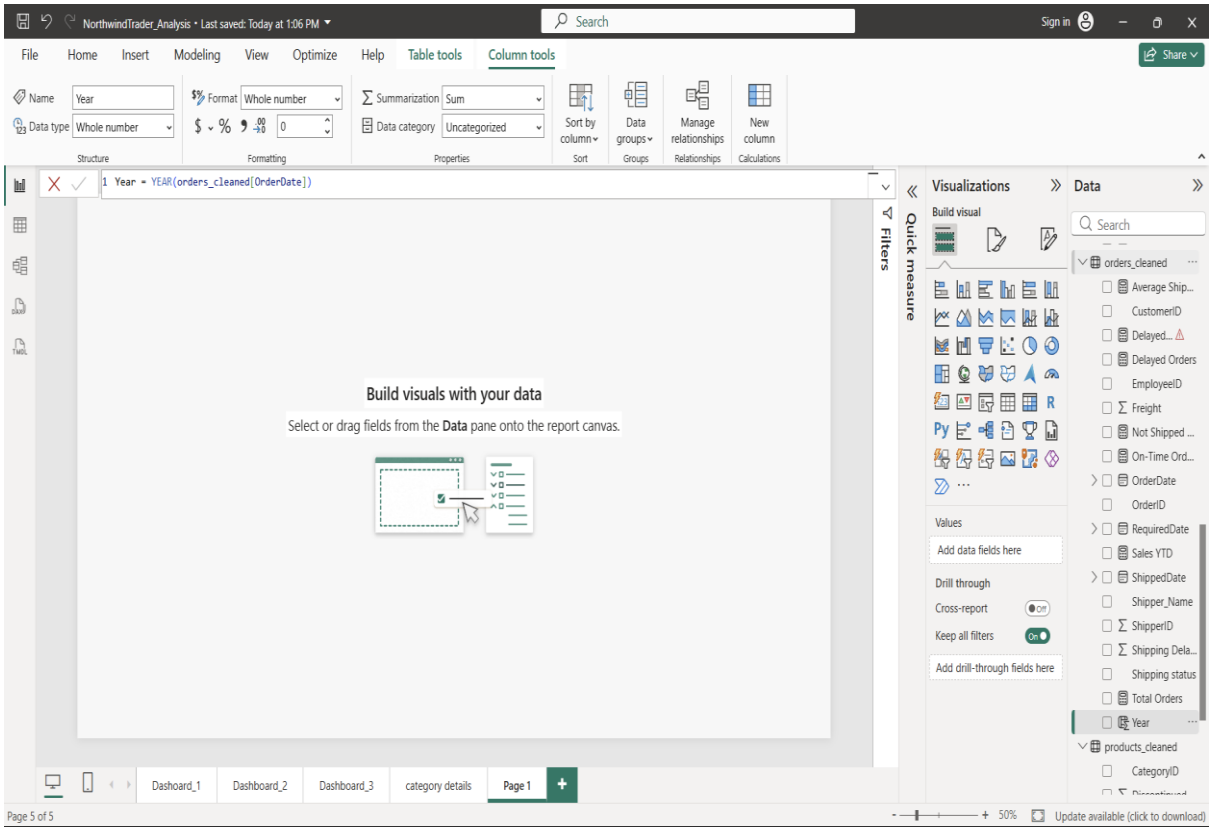
11. Created a measure for Sales YTD

```
Sales YTD =
TOTALYTD(
   [Net Sales],
   'orders_cleaned'[OrderDate]
)
```



Created New column as Year from Orders table ‘Order date’ to use in slicers.

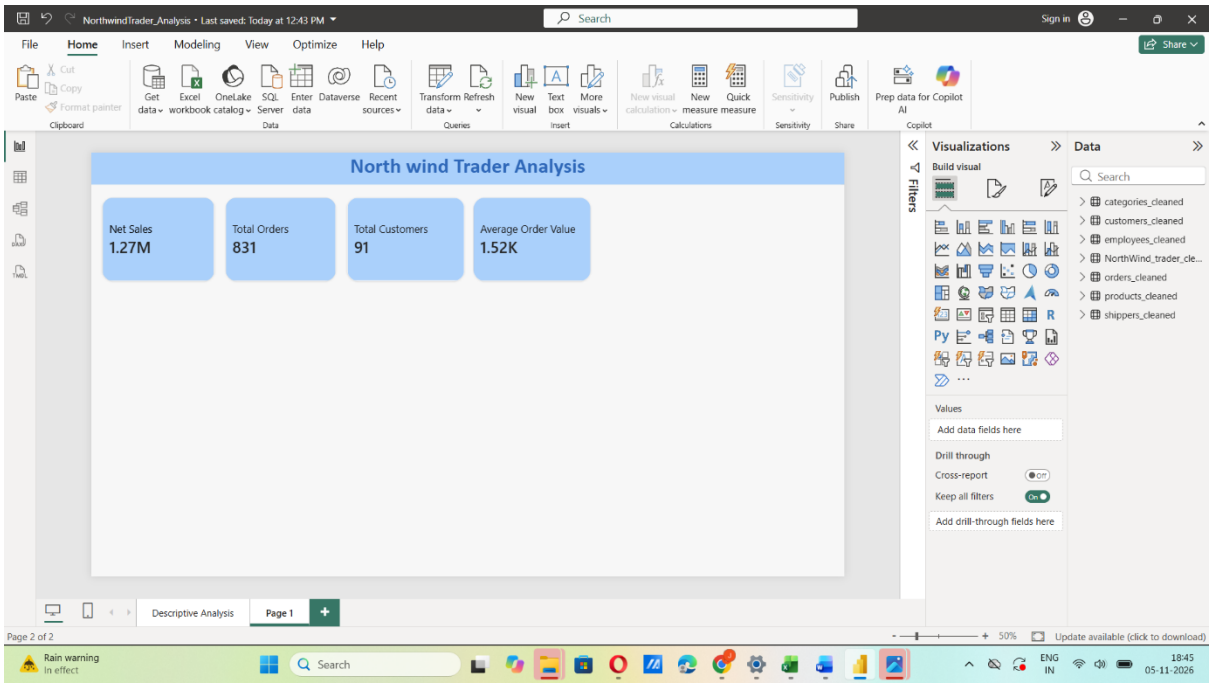
Year = YEAR(orders_cleaned[OrderDate])



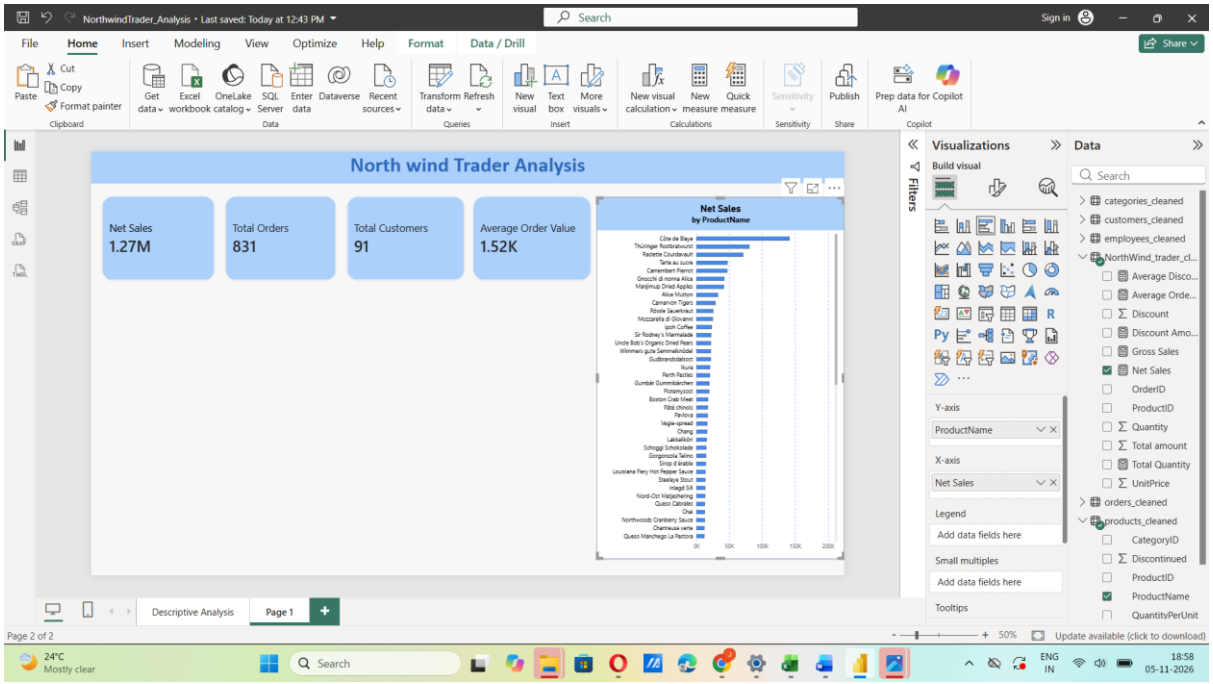
4. Visualization

1. Dashboard_1

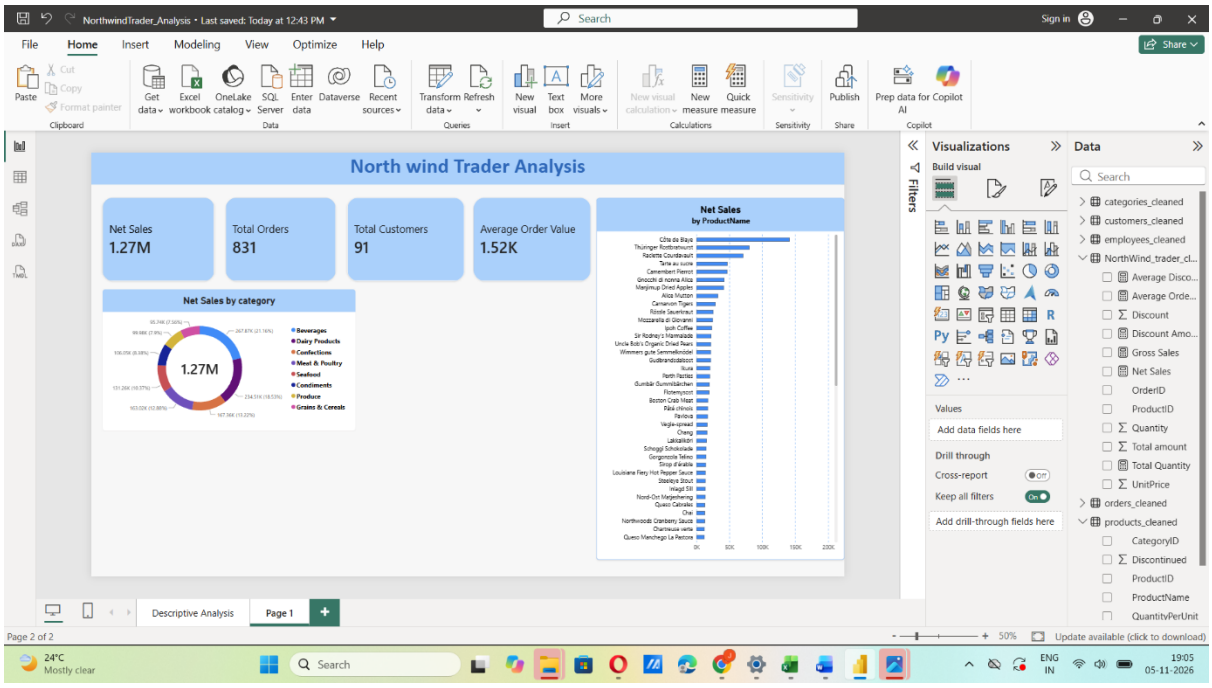
- Created Text (North wind trader Analysis) using **Text box,** changed color, adjusted font, given alignment, changed color for the text
- Created 4 card and added the Calculate measure values of **Net Sales**, **Total Orders**, **Total Customer, Average order value**.



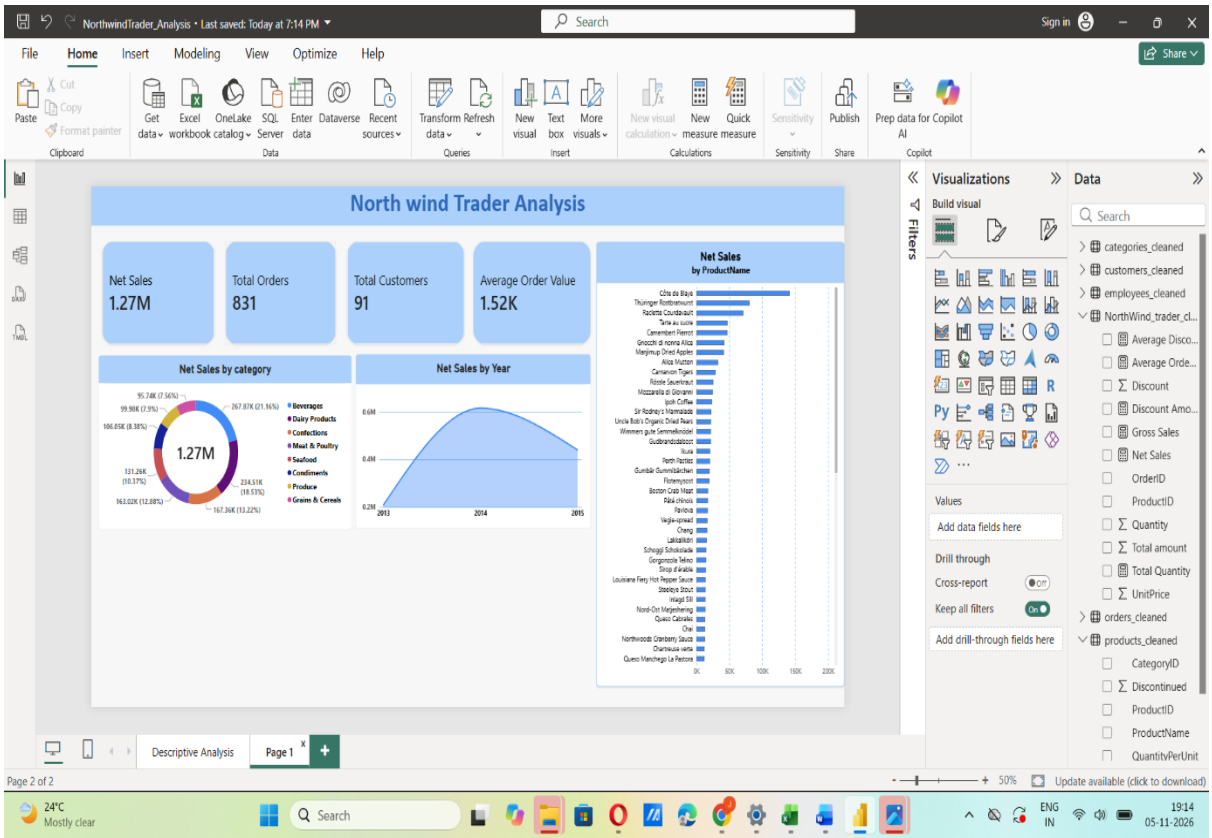
- Created a cluster Bar Chart to represent sales by Product name.
- Styled the graph for better view.



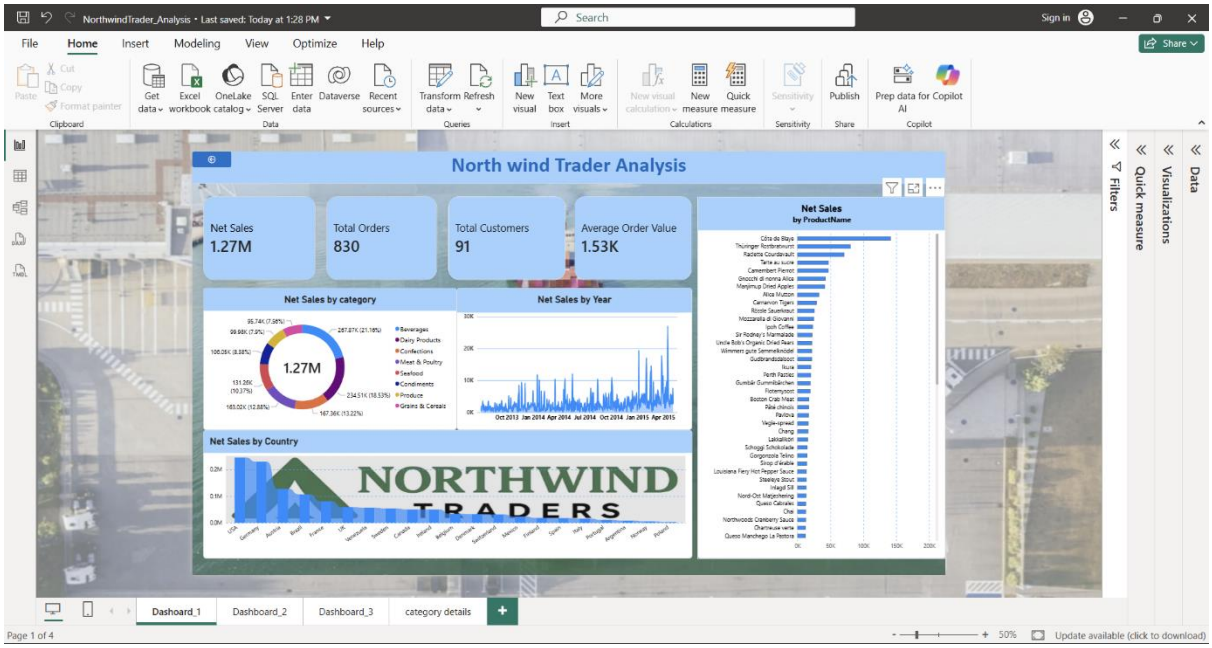
- Created donut chart to visualize Net sales by Category.
- Styled the graph for better view.



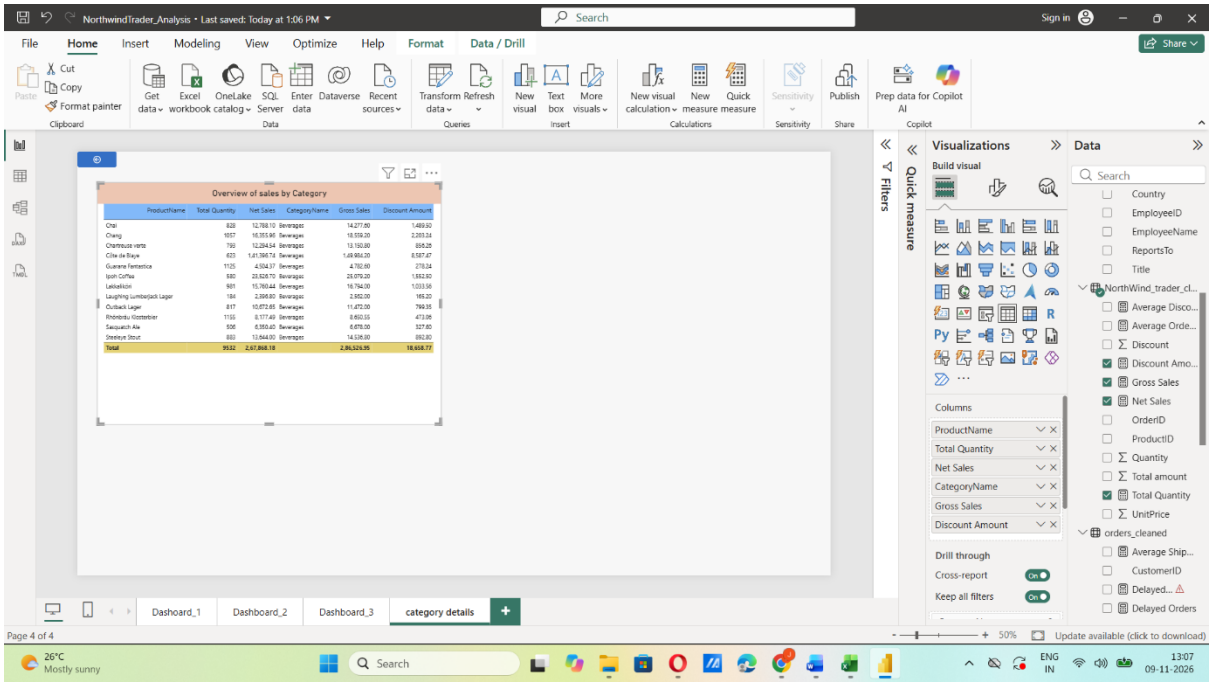
- Created line chart to Visualize Net Sales by Year.
- Styled the graph for better view.



- Created Stacked column Chart to visualize Net sales by country
- Styled the graph for better view.
- Below is complete 1 page dashboard.

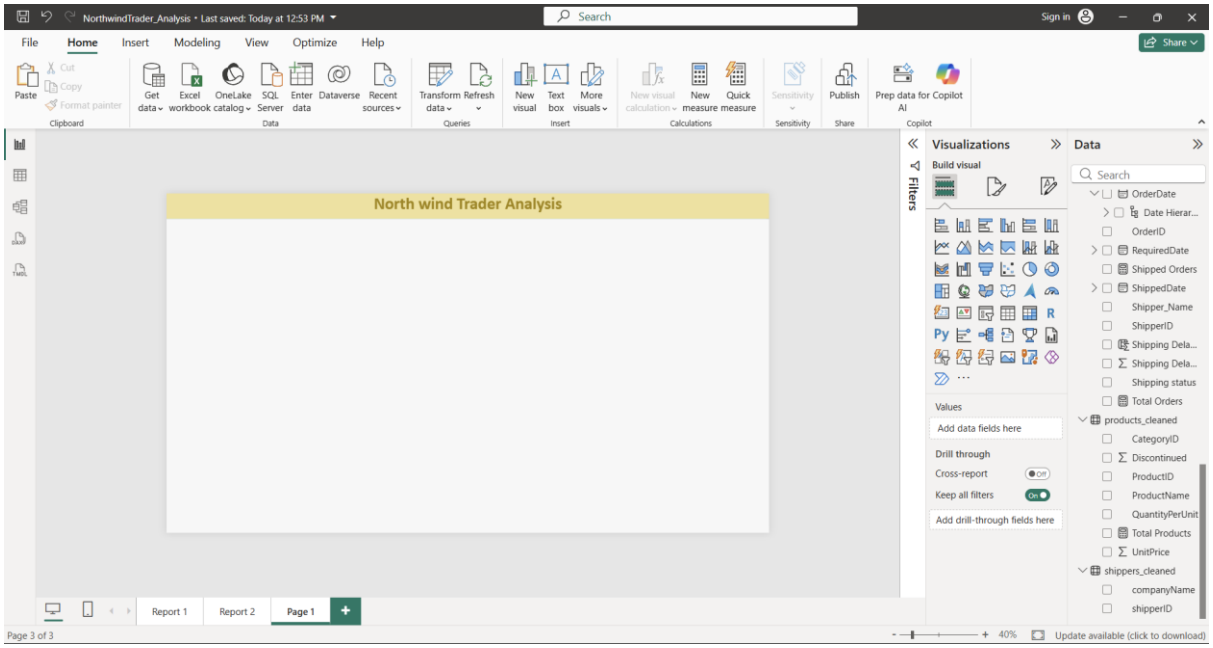


- Created DrillThrough to visualize sales by category wise when user clicks only specific category they will visualize total sales gross sales and discount .

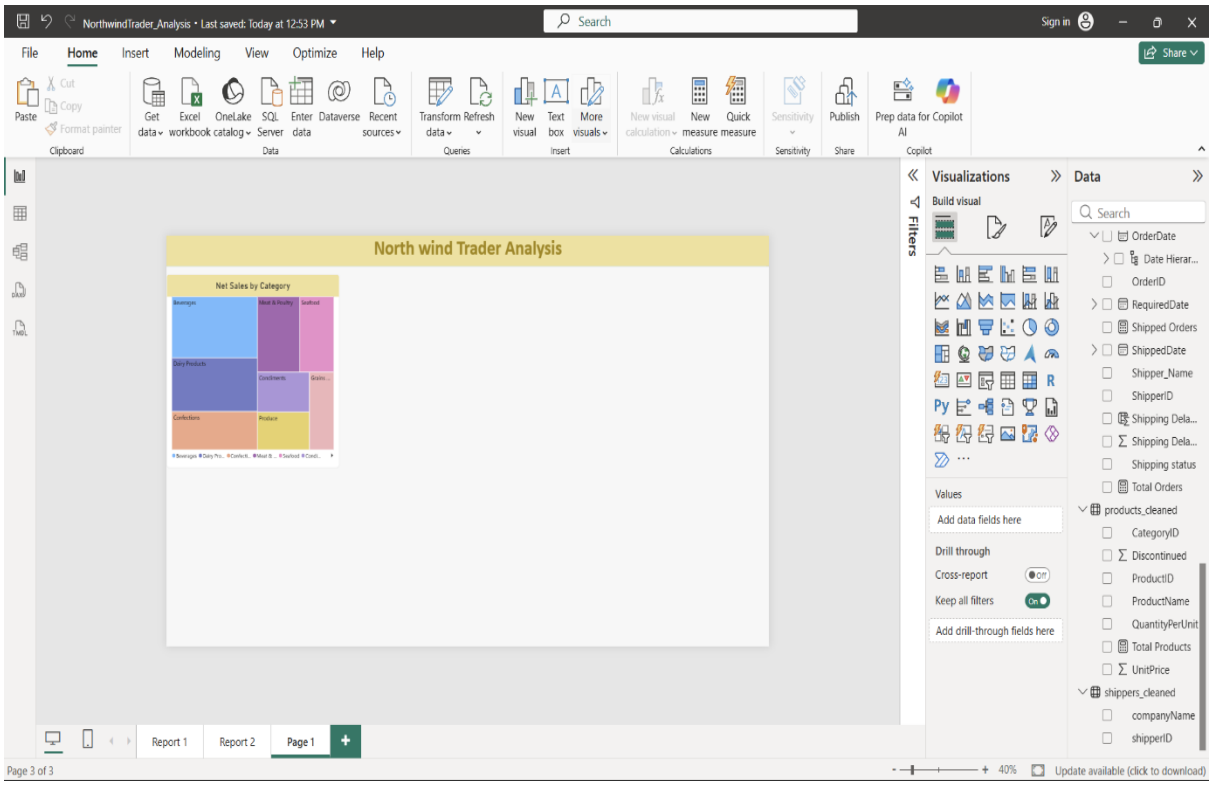


2. Dashboard_2

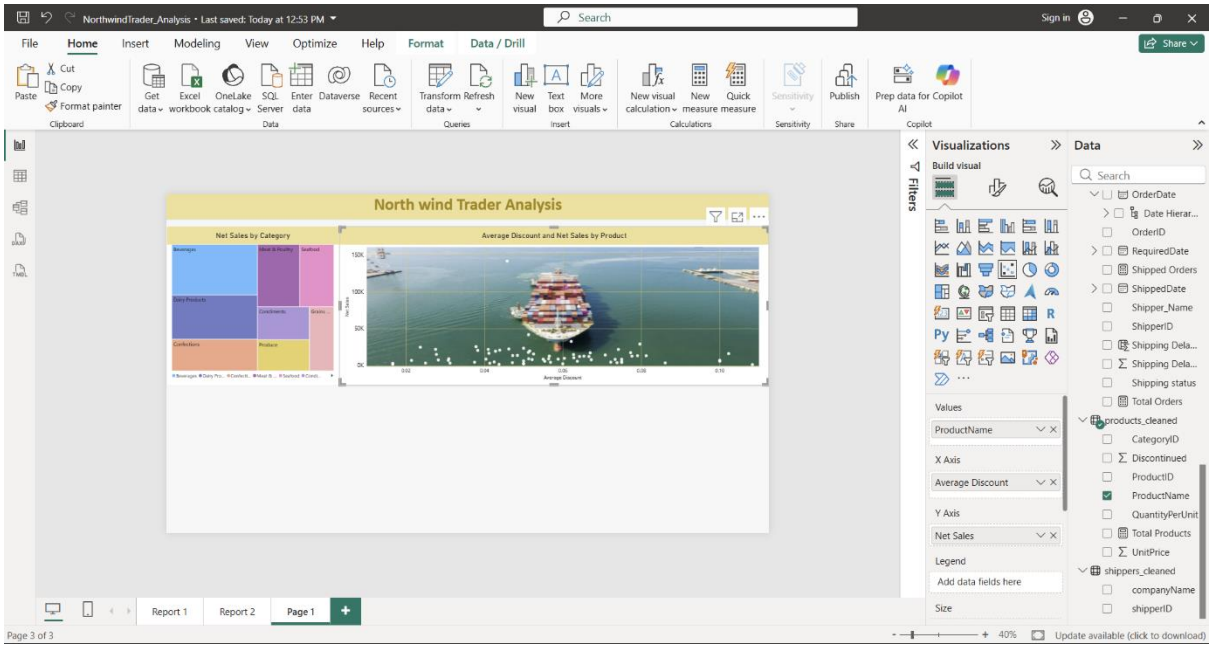
- Created Text (North wind trader Analysis) using **Text box,** changed color, adjusted font, given alignment, changed color for the text



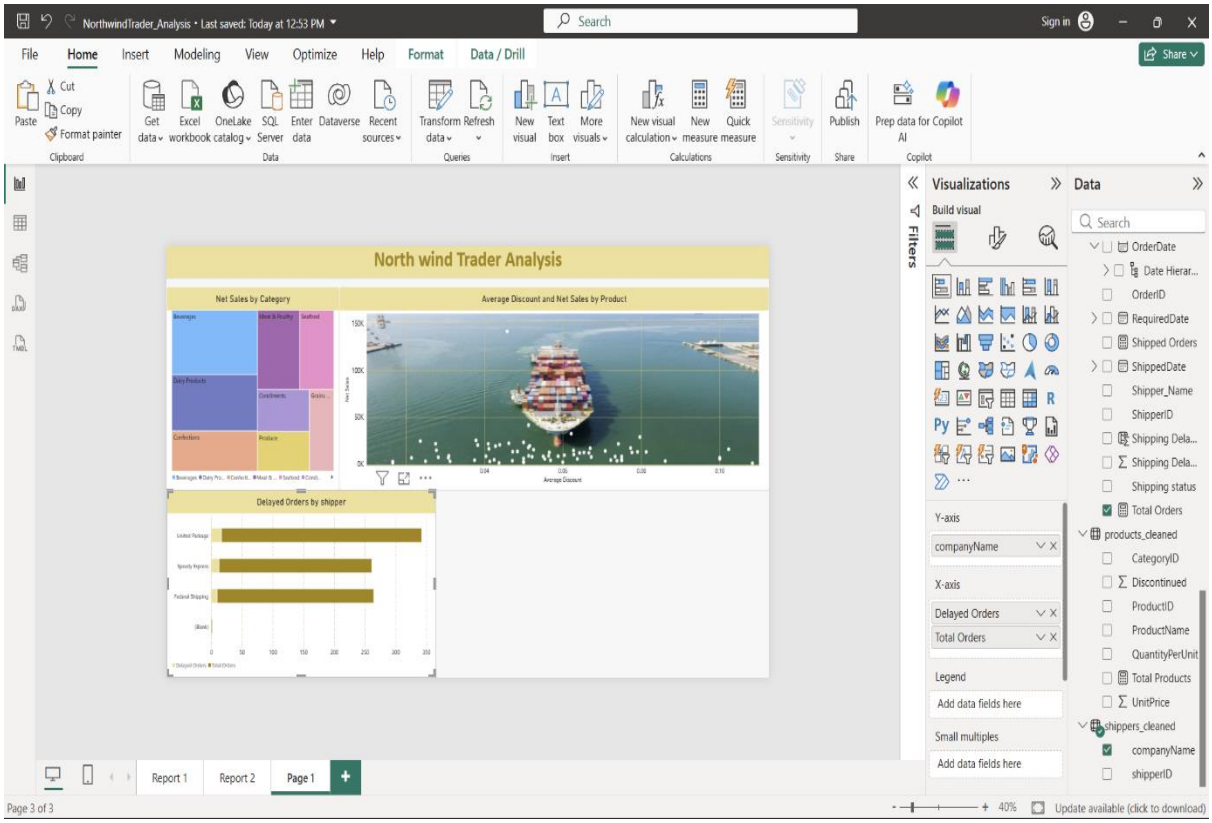
- Created a tree map chart to visualize net sales by category.
- Styled the graph for better view.



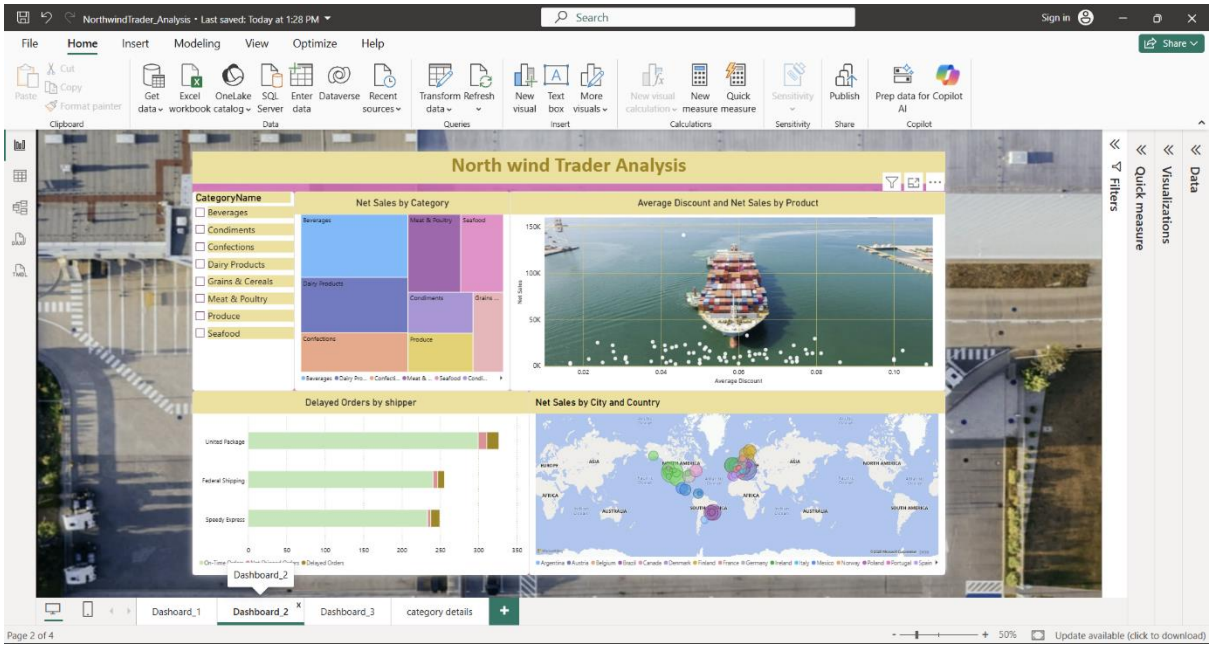
- Created scatter chart to visualize average discount and net sales by product.
- Styled the graph for better view
- Added background image for better understanding.



- Created stacked bar chart to visualize delayed orders by shipper.
- Styled graph for better view

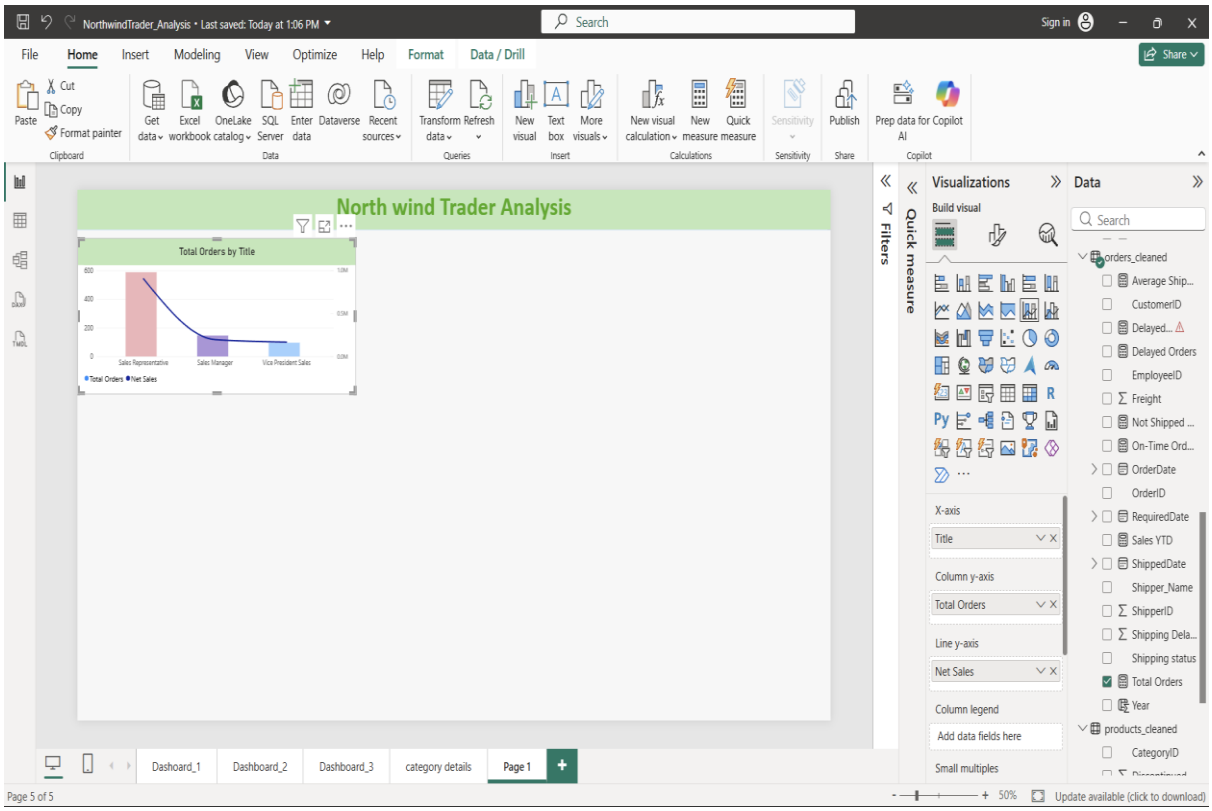


- Created Map chart to visualize net sales by city and country.
- Styled graph for better view.
- Created Slicers to visualize Category wise.

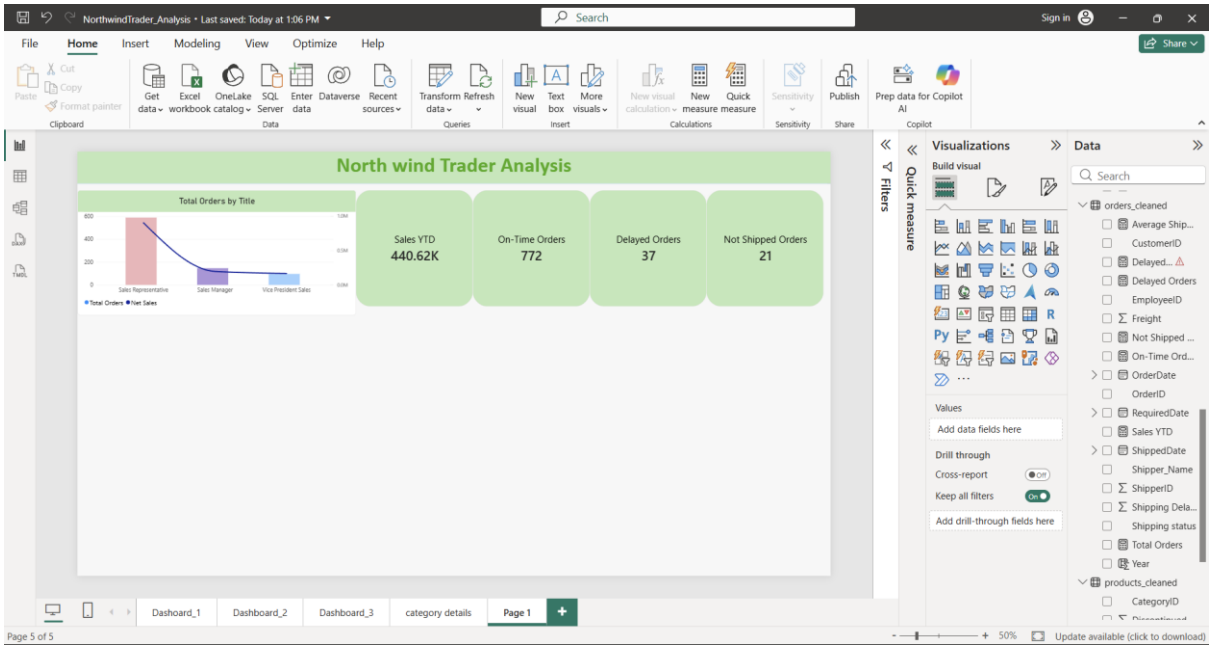


3. Dashboard_3

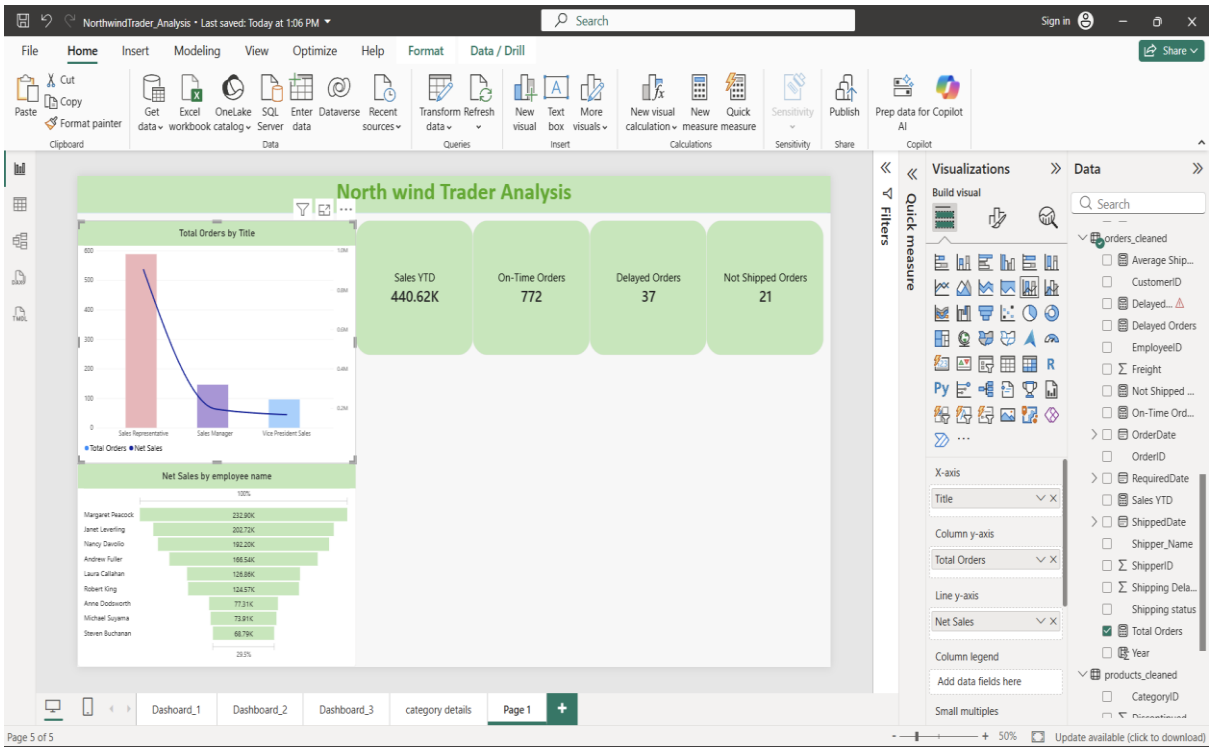
- Created a stacked bar chart to visualize total orders by title.
- Styles graph for better view.
- Line Y axis to represent net sales.



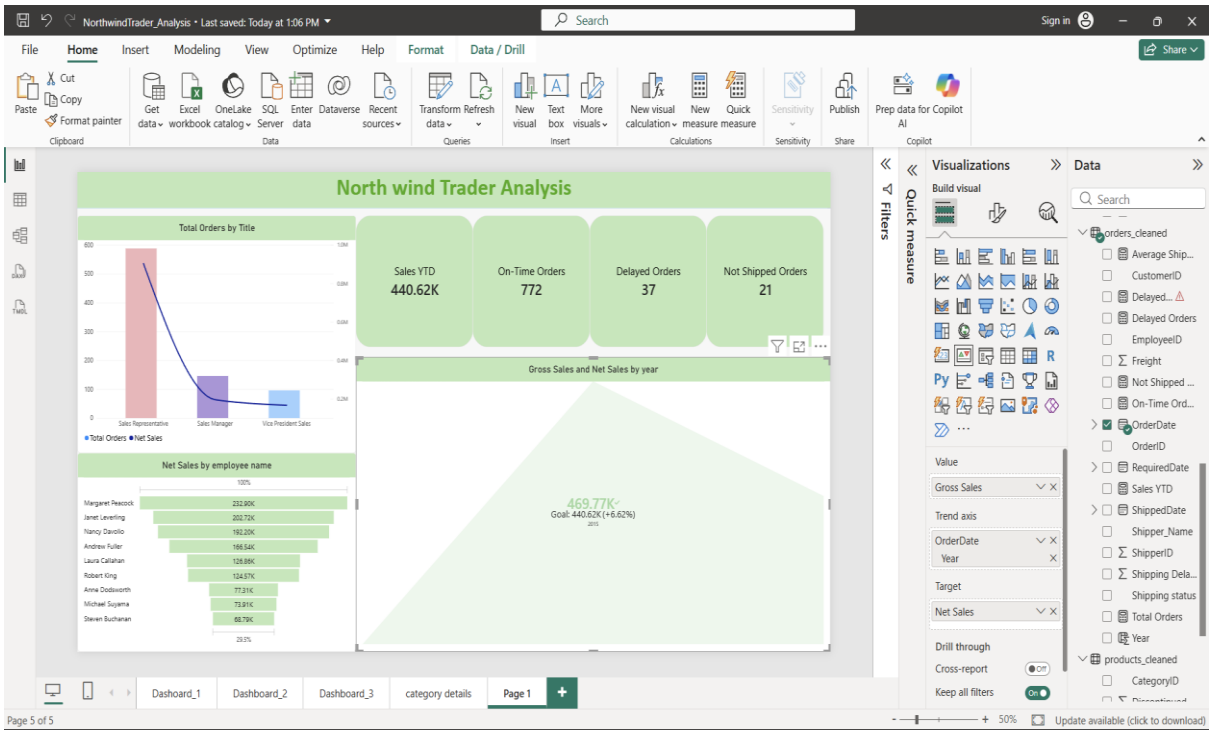
- Created 4 card to visualize values like **Sales YTD, Order shipped, Order not shipped, Delayed Order**.
- Styled graph for better view.



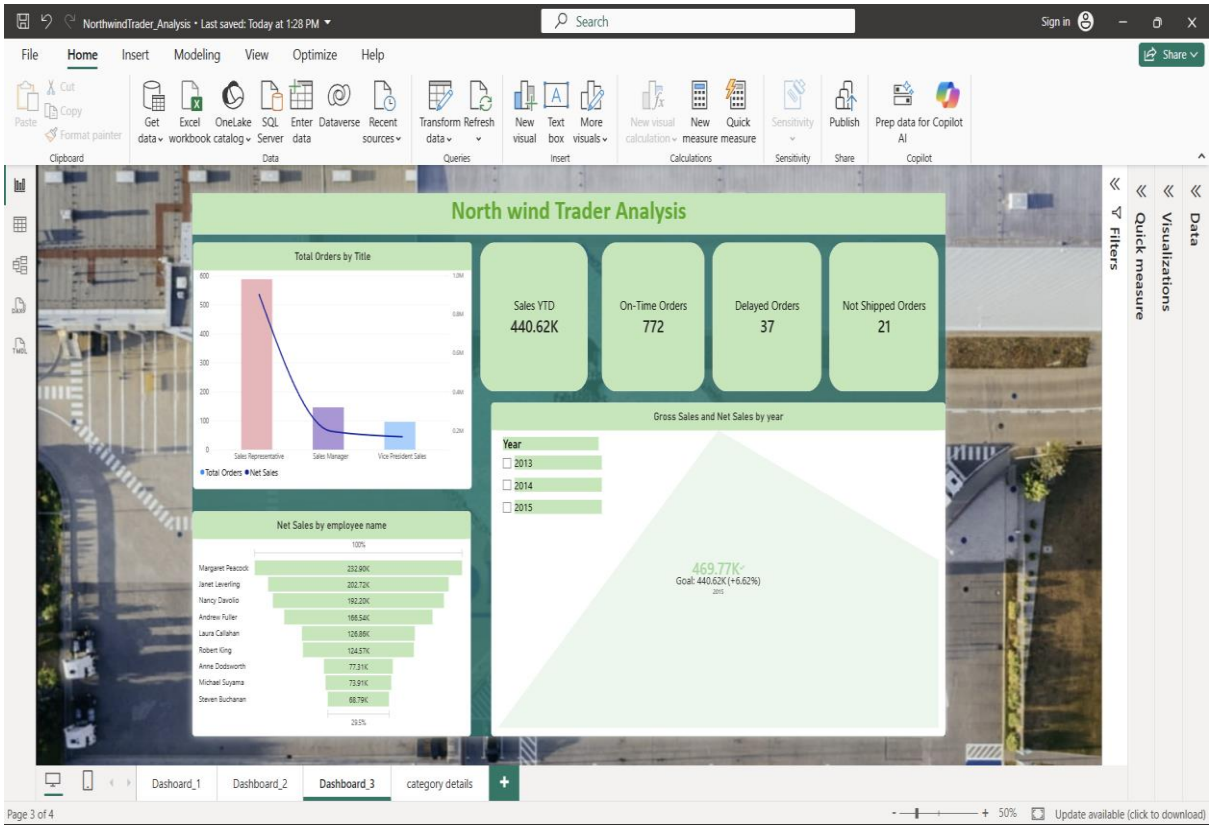
- Created Funnel chart to visualize net sales by Employee name.
- Styled graph for better view.



- Created KPI graph to visualize the target is achieved or not.
- Styled Graph for better view.



- Created slicer to visual Year wise sales trend.



- Added Bookmark and modified into button to shift from 1 dashboard to another

